

My research focuses on wave optics modeling, light shaping, and joint optimization of optical systems and reconstruction algorithms for imaging and sensing.

Computational imaging · Differentiable optics · Inverse problems · Machine learning · Diffraction · Microscopy

## Academic Appointments

|                   |                                                                                                                        |                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------|----------------------|
| 04/2024 – Present | <b>Postdoctoral Researcher</b><br>Department of Electrical and Computer Engineering, The University of Hong Kong (HKU) | Hong Kong SAR, China |
| 01/2024 – 03/2024 | <b>Research Assistant</b><br>State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)  | Hangzhou, China      |

## Education

|                   |                                                                                                                                                                                                                                        |                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 09/2018 – 12/2023 | <b>Ph.D.</b> in Optical Engineering<br>State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)<br>Dissertation: <i>Research on the principles and key issues of point spread function engineering</i> | Hangzhou, China  |
| 09/2014 – 06/2018 | <b>B.E.</b> in Optoelectronic Information Science and Engineering<br>School of Physics and Optoelectronics, South China University of Technology (SCUT)                                                                                | Guangzhou, China |

## Selected Publications

# equal contribution; \* corresponding author.

- [11] Retained accuracy with reduced precision in wave propagation modeling.  
**Xin Liu** and Yifan Peng.  
*Photonics Research*, 14(5), 1762–1774 (2026).
- [10] Learned off-aperture encoding for wide field-of-view RGBD imaging.  
Haoyu Wei, **Xin Liu**, Yuhui Liu, Qiang Fu, Wolfgang Heidrich, Edmund Y. Lam, and Yifan Peng.  
*IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 1–13 (2025).
- [9] In situ fully vectorial tomography and pupil function retrieval of tightly focused fields.  
**Xin Liu**, Shijie Tu, Yiwen Hu, Yifan Peng, Yubing Han, Cuifang Kuang, Xu Liu, and Xiang Hao.  
*Nature Communications*, 16(1), 3478 (2025).
- [8] Diffraction modeling between arbitrary non-parallel planes using angular spectrum rearrangement.  
Yiwen Hu<sup>#</sup>, **Xin Liu**<sup>#\*</sup>, Xu Liu, and Xiang Hao<sup>\*</sup>.  
*Optica*, 12(1), 39–45 (2025). [Monthly top downloads, January]
- [7] Modeling off-axis diffraction with the least-sampling angular spectrum method.  
Haoyu Wei<sup>#</sup>, **Xin Liu**<sup>#</sup>, Xiang Hao, Edmund Y. Lam, and Yifan Peng.  
*Optica*, 10(7), 959–962 (2023). [Monthly top downloads, July & August]
- [6] Fast generation of arbitrary optical focus array.  
**Xin Liu**, Yiwen Hu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao.  
*Optics and Lasers in Engineering*, 162, 107405 (2023).
- [5] Investigating deep optics model representation in affecting resolved all-in-focus image quality and depth estimation fidelity.  
**Xin Liu**<sup>#</sup>, Linpei Li<sup>#</sup>, Xu Liu, Xiang Hao, and Yifan Peng.  
*Optics Express*, 30(20), 36973–36984 (2022).
- [4] Calibration of phase-only liquid-crystal spatial light modulators by diffractogram analysis.  
**Xin Liu**, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao.  
*Optics and Lasers in Engineering*, 156, 107056 (2022).

- [3] Accurate background reduction in adaptive optical three-dimensional stimulated emission depletion nanoscopy by dynamic phase switching.  
Shijie Tu, **Xin Liu**, Difan Yuan, Wenli Tao, Yubing Han, Yan Shi, Yanghui Li, Cuifang Kuang, Xu Liu, Yufeng Yao, Yesheng Xu, and Xiang Hao.  
*ACS Photonics*, 9(12), 3863–3868 (2022). [Cover article]
- [2] Generation of arbitrary longitudinal polarization vortices by pupil function manipulation.  
**Xin Liu**, Yifan Peng, Shijie Tu, Jun Guan, Cuifang Kuang, Xu Liu, and Xiang Hao.  
*Advanced Photonics Research*, 2(1), 2000087 (2021). [Cover article]
- [1] Aberrations in structured illumination microscopy: A theoretical analysis.  
**Xin Liu**, Shijie Tu, Yan Xu, Hongya Song, Wenjie Liu, Qiulan Liu, Cuifang Kuang, Xu Liu, and Xiang Hao.  
*Frontiers in Physics*, 7, 254 (2020).

## Awards and Honors

|                  |                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026             | Outstanding Achievements Award<br>The 6th International Conference on Computational Imaging (CITA 2026)<br>Chinese Society for Optical Engineering                                       |
| 2025             | Outstanding Young Rising Star Award (Ranked 1st among 20 presenters)<br>The 5th International Conference on Computational Imaging (CITA 2025)<br>Chinese Society for Optical Engineering |
| 2024             | Honorable Mention<br>The 4th International Conference on Computational Imaging (CITA 2024)<br>Chinese Society for Optical Engineering                                                    |
| 2022             | Scholarship for Innovation and Entrepreneurship, Zhejiang University                                                                                                                     |
| 2021, 2022       | Merit Graduate Student, Zhejiang University                                                                                                                                              |
| 2020–2022        | Junshi Chen Scholarship, Zhejiang University                                                                                                                                             |
| 2020, 2021       | Outstanding Graduate Student Cadre, Zhejiang University                                                                                                                                  |
| 2019, 2021, 2022 | Outstanding Graduate Student, Zhejiang University                                                                                                                                        |

## Invited Talks

|         |                                                                                                                                                               |                      |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 09/2026 | Scaling differentiable optical simulation<br>The 6th International Conference on Computational Imaging (CITA 2026)                                            | Shenzhen, China      |
| 09/2025 | Exploring advanced imaging with computational optics<br>The 5th International Conference on Computational Imaging (CITA 2025)                                 | Suzhou, China        |
| 09/2024 | Free-space wave propagation modeling with the least-sampling angular spectrum method<br>The 4th International Conference on Computational Imaging (CITA 2024) | Xiamen, China        |
| 05/2024 | Exploring wave propagation modeling in free space<br>Guest lecture, King Abdullah University of Science and Technology                                        | Thuwal, Saudi Arabia |

## Research Mentoring

|           |                                                                                                                                       |              |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 2021–2026 | Multi-dimensional quantitative characterization methods for metasurface-modulated optical fields<br>Xinjie Sun<br>Zhejiang University | Ph.D. thesis |
| 2022–2025 | Learned wide-angle imaging with optical modeling and optimization<br>Haoyu Wei<br>The University of Hong Kong                         | Ph.D. thesis |
| 2021–2025 | Characterization of optical fields with computational methods<br>Yiwen Hu<br>Zhejiang University                                      | Ph.D. thesis |

|      |                                                                                                                                                  |                   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 2023 | Deep stereo optics imaging for high-fidelity depth estimation<br>Yuhui Liu<br>The University of Hong Kong                                        | Master's thesis   |
| 2022 | Full vectorial manipulation of tightly focused optical fields<br>Haiwei Wang<br>Zhejiang University                                              | Bachelor's thesis |
| 2022 | Numerical simulation of vectorial fields based on vectorial diffraction theory<br>Yunpeng Wang<br>Changchun University of Science and Technology | Bachelor's thesis |

## Academic Service

|                            |                                                                                                                                                                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Guest Editor</b>        | Computational Methods for Advanced Optical Imaging, Photonics, MDPI, 2026                                                                                                                                                       |
| <b>Area Chair</b>          | International Conference on Computational Photography (ICCP), 2026                                                                                                                                                              |
| <b>Program Committee</b>   | The 5th International Conference on Computational Imaging (CITA 2025)                                                                                                                                                           |
| <b>Program Chair</b>       | HKU-KAUST Joint Postgraduate Workshop on Computational Imaging, 2025                                                                                                                                                            |
| <b>Journal Reviewer</b>    | Optica; Advanced Functional Materials; Light: Advanced Manufacturing; Laser & Photonics Reviews; Advanced Science; Optics Letters; Optics Express; JOSA A; JOSA B; Applied Optics; Journal of Microscopy; Optics Communications |
| <b>Conference Reviewer</b> | SIGGRAPH; SIGGRAPH Asia; ICCP                                                                                                                                                                                                   |

## Patents

|      |                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | A method and device for calibration of phase-only spatial light modulators<br>Xiang Hao, <b>Xin Liu</b> , Yubing Han, Cuifang Kuang, and Xu Liu<br>Granted patent: CN112697401B  |
| 2020 | A method for point spread function reconstruction<br>Xiang Hao, <b>Xin Liu</b> , Cuifang Kuang, and Xu Liu<br>Granted patent: CN110133849B                                       |
| 2020 | A two-dimensional rapid beam scanning module for confocal fluorescence microscopy<br>Xiang Hao, <b>Xin Liu</b> , Shijie Tu, and Xu Liu<br>Granted patent: CN109491065B           |
| 2020 | A device for common-path beam modulation in imaging and lithography systems<br>Xu Liu, <b>Xin Liu</b> , Xiang Hao, Cuifang Kuang, and Haifeng Li<br>Granted patent: CN110568650B |
| 2020 | A device for generating super-resolution optical focus arrays<br>Xiang Hao, <b>Xin Liu</b> , Cuifang Kuang, Xu Liu, and Haifeng Li<br>Granted patent: CN110568731B               |
| 2020 | A multicolor super-resolution microscope with automatic alignment capability<br>Xiang Hao, Shijie Tu, <b>Xin Liu</b> , and Xu Liu<br>Granted patent: CN109358030B                |